

- **AI-GENERATED REPORT** — see the note on methodology on the final page

## PROGRAMME ASSESSMENT

# IndiaFOSS 2026 Conference Programme Analysis

A structural assessment of the scheduled conference programme — what it covers well, where it thins out, and what a wholesome FOSS & digital-commons programme would still need.

**137**

SCHEDULED  
SESSIONS

**68.3h**

PROGRAMME  
MINUTES

**10**

TRACKS /  
DEVROOMS

**3**

CONFERENCE  
DAYS

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Source: if2026-schedule.ods (General Track + 8 devrooms) plus the 8 pre-conference Workshop proposals from cfp-all.csv (approx. 2 hours each, held the day before the main conference), cross-referenced against the published CFP proposals at [fossunited.org/c/indiafoss/2026/cfp](https://fossunited.org/c/indiafoss/2026/cfp). This report, including every session's classification, was generated by an AI assistant (Claude) — it is not an official FOSS United publication. Treat it as an analytical starting point, not a substitute for programme-committee judgement; see the methodology note on the final page for how the classification was done and where it is most likely to be wrong.

## Executive Summary

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IndiaFOSS 2026's programme is **large, technically serious, and unusually community-minded**. Counting the pre-conference workshop day, the conference runs **137 sessions across roughly 68 hours** over three days, and builds its identity around four consistent pillars: a deep **systems/embedded engineering** cluster (Android internals, RTOS, compilers, open hardware, security), a broad **FOSS community & civic-commons** strand running through the General Track (contributor culture, GovTech, digital public goods, policy), a full dedicated **Documentation & Technical Writing** devroom — a track most conferences of this size skip entirely — and a **pre-conference workshop day** that gives eight topics a genuinely hands-on, 2-hour treatment. Average technical depth (5.7/10) and novelty (5.6/10) sit in a healthy "practitioner-intermediate" band: this is a conference of people telling war stories from real, shipped systems, not a research symposium and not a beginner bootcamp.

The programme is also **lopsided in three predictable ways**: within the two main conference days, format skews heavily toward short exposure (43% of all sessions are 15-minute-or-under lightning talks) at the expense of sustained depth; content skews toward the **application layer** even inside "systems" tracks; and the eight workshops, while a genuine strength, sit thematically apart from the conference's own technical core — none of them extend the RTOS, AOSP, Hardware, Compilers or Security devrooms into hands-on territory. Several other large, active areas of the FOSS/digital-commons world — data engineering & ML systems, accessibility, and full-stack open hardware beyond firmware — remain present only as a trace rather than as first-class content.

### Gap analysis — at a glance

#### • Well represented

- **Embedded systems depth** — AOSP + RTOS + Compilers + Hardware + Security together contribute 715 minutes of high-depth content (avg. depth 6.4–7.6/10), the technical core of the conference.
- **FOSS community & contribution culture** — 74 of 137 sessions touch "Community" as a domain; onboarding, maintainership, and governance are recurring threads.
- **Documentation as a first-class discipline** — a full 11-session, 150-minute devroom, rare at this scale.
- **A genuine hands-on workshop day** — 8 pre-conference workshops (2 hours each, 960 minutes total) on Qt, OCaml, generative art/live-coding, AI agents, OpenStreetMap mapping, self-hosting, and digital security give real skill-building a dedicated day rather than a token slot.
- **Civic tech / GovTech / digital-commons data** — judiciary mapping, GovTech archives, water intelligence, census/data-for-India work form a

#### • Thin or missing

- **Workshops don't reach the conference's own technical core** — the 8 hands-on sessions skew toward general-purpose software, creative-coding and civic themes (Qt, OCaml, strudel.cc, p5.js art, AI agents, OSM, self-hosting, digital security); none is a hands-on extension of RTOS, AOSP, Hardware, Compilers or Security — the very devrooms with the highest average depth (6.4–7.6/10) get zero hands-on treatment.
- **Data engineering & ML/AI systems have no home track** — AI/agent themes surface in roughly a quarter of the programme by minutes but are scattered one-off mentions inside Cloud & DevOps, General Track, Compilers, and now one workshop, rather than a coherent, curated strand.
- **Open hardware beyond firmware is the smallest devroom** — only 6 sessions / 175 minutes, dwarfed by the software-adjacent embedded tracks (RTOS 10, AOSP 11). PCB

coherent public-interest thread inside the General Track.

- **Compiler & PL depth** — LLVM, formal verification, Rust, Julia, WASM: a genuine deep-tech devroom.

design, manufacturing, and hardware supply-chain topics are nearly absent.

- **Policy & legal depth is shallow** — 13 sessions touch Legal/Policy, but almost all are 15-minute lightning talks or BoFs; software patents, age-verification law, and content regulation get exposure, not sustained treatment.
- **Accessibility, localisation and inclusive design** appear only incidentally (one lightning talk on FOSS accessibility, one on Tamil translation) rather than as a named thread.
- **Sustainability/business-model-of-FOSS content is discussion-only** — pay competitiveness, cooperatives, and monetisation are raised in BoFs (60-minute, unstructured) rather than researched talks.

### Net assessment

The programme is **wholesome for a systems-and-community-minded FOSS audience** — a software developer, embedded engineer, or community organiser will find a full three days of relevant, well-paced content, including genuine hands-on time on the workshop day. It is **less wholesome for a data/ML practitioner, a hardware maker beyond firmware, or a policy specialist wanting depth** — and the one attendee still under-served on the skill-building front is exactly the person the deep-tech devrooms attract: someone who came for RTOS, AOSP, Hardware, Compilers or Security gets zero hands-on options in their own area, even though the workshop day proves the organisers know how to run one.

# Detailed Overview

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## 1. How the programme is built

The schedule has three layers. A **pre-conference Workshop day** (8 sessions, 960 minutes) runs the day before the main event, each workshop given a full 2-hour hands-on slot. Then, across the two main conference days, the **General Track** (54 sessions, 1,800 minutes — 44% of all programme time) runs across three halls and rooms and is the conference's broadest surface: general FOSS talks, an invited talk, two panel discussions, and eight BoFs sit alongside standard 30-minute talks. The remaining eight sheets are **half-day mini-conferences ("devrooms")** — AOSP, Compilers, Documentation & Technical Writing, Cloud & DevOps, RTOS, Security, Design, and Hardware — each concentrated into a single afternoon or morning block on one day, in a single room. Devrooms together contribute 75 sessions and 1,340 minutes (33%). This structure means a single attendee physically cannot sample more than a handful of devrooms across the two days, so the devroom line-up functions less like "an even split" and more like eight parallel mini-conferences that different sub-communities self-select into — with the workshop day functioning as a distinct, opt-in fourth event of its own.

## 2. Format and duration: breadth over depth on the main days

Of the 129 sessions across the two main conference days, **59 (46%) are lightning talks of 15 minutes or less**, and **56 (43%) are standard 20–30 minute talks**. Only 12 sessions run 60 minutes or longer, and those are almost entirely BoFs and panel discussions (open-floor formats), not deep single-speaker technical dives — there is effectively no 60-minute solo deep-technical talk slot anywhere in the two-day programme. Two devrooms — **Cloud & DevOps and RTOS** — are **90% lightning talks**, meaning attendees get rapid exposure to many ideas (10 topics in roughly 2.5 hours each) but little room for any single idea to be explored past its motivating problem. By contrast, **Hardware and Design run zero lightning talks** — every session there gets at least 20 minutes, and it shows in the narrative completeness of those proposals. This is a real editorial trade-off already made by the programme committee: devrooms with the most novel/frontier material (RTOS, avg. novelty 6.5/10) are also the ones giving each idea the least time to land. The workshop day is the counterweight to all of this: its 8 sessions are uniformly 120 minutes, the longest, most sustained format anywhere in the conference.

## 3. What kind of work is this? Overall category and stack level

By minutes across all 137 sessions, the programme splits **Software 2,030 min (50%) · FOSS Philosophy/Principles 1,220 min (30%) · Data 520 min (13%) · Hardware 330 min (8%)**. Nearly a third of scheduled time goes to community, governance, culture, policy and career topics rather than shipped technical artefacts — a signal that IndiaFOSS positions itself as a movement-building conference as much as a technical one. Within the technical stack-level tags, **"Application" dominates at 91 tag-instances**, far ahead of Frameworks/Middleware (20), Firmware (17), Operating System (16), Libraries (12), Hardware (9) and Kernel (7) — even the RTOS and AOSP devrooms, nominally "systems" tracks, spend much of their time at the firmware/application boundary (bootloaders, HALs, drivers) rather than deep kernel or silicon work. Genuine kernel- and silicon-level content (RISC-V crypto acceleration, TriCore bring-up, HackerFab semiconductor fabrication, sandbox kernels) exists but is a minority inside a minority, and the workshop day — being introductory by design — adds to the application-layer weight rather than the kernel/hardware end.

## 4. Domains served

Tagged against the domains in *categories.md*, sessions cluster heavily around **Community (74)** and **Industry (57)**, with **Research (27)**, **Education/Academia (19)**, **Not-for-Profit (17)**, **Government (14)** and **Legal/Policy (13)** trailing well behind (sessions can carry more than one domain tag). Government and civic-tech content is real and often high-quality — the Telangana Oracle-to-PostgreSQL migration, the NHAI AI-workflow talk, the Judiciary Entity Map, the CBSE exam-security disclosure story — but at 14 tagged sessions it is a minor thread relative to the scale of India's GovTech/DPI ecosystem that FOSS United's own General-Track panel ("From DPI to People's Digital Infrastructure") explicitly calls out as underserved. Education/Academia content gets a boost from the workshop day (Qt, OCaml and generative-art workshops are explicitly pitched at learners) but is still sparse given how much of the FOSS pipeline (GSoC, campus communities, COSH-style centres) depends on it.

## 5. Depth vs. novelty, and who the programme is for

Across all 137 sessions, average depth is **5.7/10** and average novelty **5.6/10**, with RTOS (7.6 depth) and Security (6.8 depth) the deepest tracks, and the Workshop day (4.6 depth, by design — these are onboarding sessions for newcomers) and Documentation & Technical Writing (4.5 depth) intentionally the most accessible. The dashboard below breaks depth vs. novelty out by overall category: Hardware sessions cluster at high depth/high novelty (little hand-holding, aimed at practitioners already in the space), while FOSS Philosophy/Principles sessions cluster at low-to-mid depth (accessible almost by design, since the goal is persuasion and culture-building rather than technique transfer). Audience tags mirror this: **software developers are the explicit or implicit audience for 98 of 137 sessions**, with professionals (52), researchers (38) and hackers (37) next; educators/professors (13) and designers (10) are the least served named audiences from the *categories.md* list, and venture-capitalist-oriented content is essentially absent (business people appear only tangentially, in 13 sessions, none of them addressed primarily to a funding/investment audience).

# IndiaFOSS 2026 — Programme at a Glance

137 scheduled sessions (incl. pre-conference workshops) · 4,100 minutes · 10 tracks

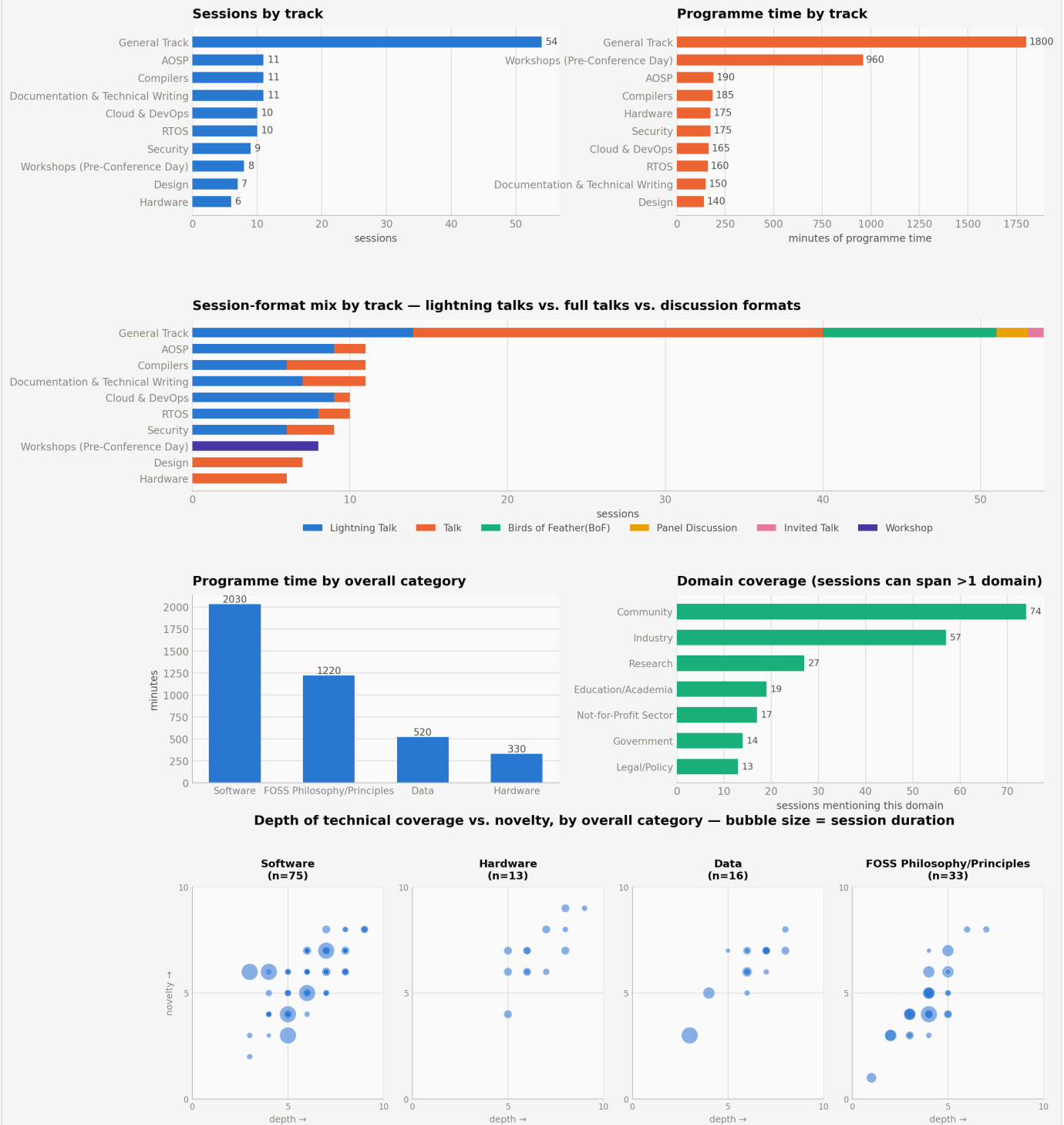


Figure 1. Programme composition across the six metrics tracked in this assessment: session and time distribution by track (including the pre-conference Workshop day), format mix, overall-category time split, domain coverage, and the depth/novelty relationship by category (bubble size = session duration in minutes).

## 6. The ambient-AI observation

No track or devroom is named for AI or machine learning, yet AI/agent-related themes surface in roughly a quarter of the programme by scheduled minutes — sandboxing AI agents, AI on-call agents, KV-cache systems work, AI-driven language assessment, AI evaluation frameworks for non-profits, one full hands-on

workshop on building AI agents, and multiple policy sessions on AI-generated imagery and open-weight model harms. This is arguably the right editorial call for a conference about *Free and Open Source Software* rather than AI specifically — but it also means attendees looking for a coherent "open/local AI systems" thread have to assemble it themselves from talks and one workshop scattered across five unrelated tracks.

## 7. The pre-conference workshops, in detail

The 8 workshops are: *Getting Started with Qt* (desktop GUI development), *Making Music with strudel.cc* (live-coding, complete-beginner friendly), *Digital Security for Journalists, Activists, and Everyday Internet Users* (privacy/threat-modelling), *Fun and Profit with OCaml* (functional programming, run by IIT Madras's FP Launchpad), *Art – meet Math – meet Code* (generative art with p5.js), *Building AI Agents End-to-End* (agent frameworks, MCP, observability), *Adding Places to OpenStreetMap* (an outdoor mapping session), and *Self-Hosting 101* (homelab/server setup). Read as a set, this is a genuinely well-rounded introductory slate — it spans desktop software, functional programming, civic mapping, digital rights, creative coding, and modern AI tooling, and three of the eight (strudel.cc, Art/Math/Code, OpenStreetMap) are explicitly designed for participants with no prior programming background, which is rare generosity in a conference this technical. The gap is thematic rather than qualitative: an attendee who wants to walk in the door already excited about RTOS, embedded Android, compilers, open hardware or security has no equivalent hands-on on-ramp — every one of those devrooms remains talk-only, even though the organisers have clearly demonstrated, via the other eight workshops, that they can run a hands-on session well.

# Full Session Metrics Table

All 129 scheduled sessions, classified against the *categories.md* schema: Platform, Domain, Level in Technology Stack, Overall category, Depth of technical coverage (1–10), Novelty vs. existing work (1–10), and Suitability for audience. Sessions are grouped by track, then by scheduled time. Title links to the original CFP proposal on fossunited.org. Multi-value fields are separated by “;”.

| Track          | Session                                                                                                                 | When                             | Min | Format         | Platform        | Domain                                   | Stack Level                            | Category | Depth | Novelty | Audience                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----|----------------|-----------------|------------------------------------------|----------------------------------------|----------|-------|---------|----------------------------------------------------------------|
| AOSP           | <a href="#">Keeping up with the AOSP</a>                                                                                | Day 1 (26 Sep)<br>Room 1 · 14:00 | 10  | Lightning Talk | Mobile          | Community;Industry                       | Operating System                       | Software | 4     | 3       | software developers, professionals                             |
| AOSP           | <a href="#">Under the Hood of Android , What Really Happens When You Open an App?</a>                                   | Day 1 (26 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk | Mobile          | Education/Academia;Community             | Operating System                       | Software | 5     | 4       | software developers, graduate/master students                  |
| AOSP           | <a href="#">Launcher3 Internals: Understanding Android's Home Screen, Google Feed Integration, and Custom -1 Panels</a> | Day 1 (26 Sep)<br>Room 1 · 14:00 | 25  | Lightning Talk | Mobile          | Community;Industry                       | Application                            | Software | 6     | 6       | software developers, professionals                             |
| AOSP           | <a href="#">Inside Android Keyboards: Building an Open Source IME</a>                                                   | Day 1 (26 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk | Mobile          | Community                                | Application;Frameworks/Middleware      | Software | 6     | 6       | software developers, hackers                                   |
| AOSP           | <a href="#">Adapting Generic Boot Loader on Android Platforms</a>                                                       | Day 1 (26 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk | Embedded;Mobile | Industry                                 | Firmware;Operating System              | Software | 8     | 7       | software developers, professionals                             |
| AOSP           | <a href="#">Porting AOSP to Allwinner Sunxi platform</a>                                                                | Day 1 (26 Sep)<br>Room 1 · 14:00 | 25  | Talk           | Embedded;Mobile | Community;Industry                       | Kernel;Operating System                | Software | 9     | 8       | software developers, hackers, professionals                    |
| AOSP           | <a href="#">Living With a DeGoogled Phone: Privacy, Trade-offs, and Daily Use</a>                                       | Day 1 (26 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk | Mobile          | Community                                | Application                            | Software | 3     | 3       | hackers, social workers                                        |
| AOSP           | <a href="#">Volume Controller using Knob (Potentiometer)</a>                                                            | Day 1 (26 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk | Embedded;Mobile | Community;Education/Academia             | Hardware;Firmware                      | Hardware | 6     | 6       | hackers, graduate/master students                              |
| AOSP           | <a href="#">Zero Trust, Zero TEE: Porting Android 16 to a Device Whose Security Stack Doesn't Exist Yet</a>             | Day 1 (26 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk | Mobile          | Community;Industry                       | Kernel;Operating System                | Software | 9     | 8       | software developers, professionals, hackers                    |
| AOSP           | <a href="#">Beyond HIDL: Modernizing Android HALs with Rust &amp; Stable AIDL</a>                                       | Day 1 (26 Sep)<br>Room 1 · 14:00 | 25  | Talk           | Mobile          | Industry                                 | Operating System;Frameworks/Middleware | Software | 8     | 6       | software developers, professionals                             |
| AOSP           | <a href="#">Bypassing Android MTP: pushing a native C++ daemon via ADB for fast storage analysis</a>                    | Day 1 (26 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk | Mobile;Desktop  | Community                                | Application                            | Software | 6     | 7       | software developers, hackers                                   |
| Cloud & DevOps | <a href="#">How OpenTelemetry Instrumentation Works Under the Hood</a>                                                  | Day 2 (27 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cloud/Server    | Industry                                 | Libraries;Frameworks/Middleware        | Software | 7     | 5       | software developers, professionals                             |
| Cloud & DevOps | <a href="#">Remote Execution by Design: Sandboxing AI Agents with gVisor &amp; Pipelock</a>                             | Day 2 (27 Sep)<br>Hall 3 · 14:00 | 30  | Talk           | Cloud/Server    | Industry;Research                        | Kernel;Operating System                | Software | 8     | 7       | software developers, professionals, researchers                |
| Cloud & DevOps | <a href="#">From Prompts to Production: Building an Open-Source AI On-Call Agent with MCP and Multi-Agent Workflows</a> | Day 2 (27 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cloud/Server    | Industry                                 | Frameworks/Middleware;Application      | Software | 6     | 5       | software developers, professionals                             |
| Cloud & DevOps | <a href="#">Long live outbox : Outbox is dead</a>                                                                       | Day 2 (27 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cloud/Server    | Industry                                 | Libraries;Application                  | Software | 7     | 6       | software developers, professionals                             |
| Cloud & DevOps | <a href="#">Kubernetes Native Observability Dashboards with Perseus, without the YAML/JSON Hell</a>                     | Day 2 (27 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cloud/Server    | Industry;Community                       | Frameworks/Middleware;Application      | Software | 6     | 6       | software developers, professionals                             |
| Cloud & DevOps | <a href="#">Cilium Network Policy Anti-Patterns I Learned the Hard Way</a>                                              | Day 2 (27 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cloud/Server    | Industry                                 | Frameworks/Middleware                  | Software | 6     | 4       | software developers, professionals                             |
| Cloud & DevOps | <a href="#">Escaping Cron Hell: Building and Contributing to Dagou, an Open Source Workflow Orchestrator</a>            | Day 2 (27 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cloud/Server    | Community;Industry                       | Application                            | Software | 5     | 5       | software developers, professionals                             |
| Cloud & DevOps | <a href="#">15 Years, One Server, 8GB RAM and 500,000 Linux Learners: Lessons from Running Webminal</a>                 | Day 2 (27 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Web             | Education/Academia;Not-for-Profit Sector | Application                            | Software | 4     | 6       | educators/professors, graduate/master students, social workers |
| Cloud & DevOps | <a href="#">Unused Is Not a Kubernetes Field: Platform Hygiene Is Not Cleanup</a>                                       | Day 2 (27 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cloud/Server    | Industry                                 | Application                            | Software | 5     | 4       | software developers, professionals                             |
| Cloud & DevOps | <a href="#">I Deleted Production With helm upgrade. Here's the Exact Chain of Events.</a>                               | Day 2 (27 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cloud/Server    | Industry                                 | Application                            | Software | 5     | 4       | software developers, professionals                             |

| Track                             | Session                                                                                                   | When                             | Min | Format         | Platform               | Domain                             | Stack Level                           | Category                          | Depth | Novelt<br>y | Audience                                                          |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------|-----|----------------|------------------------|------------------------------------|---------------------------------------|-----------------------------------|-------|-------------|-------------------------------------------------------------------|
| Compilers                         | Semantic Quantization in KV Cache Eviction                                                                | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cross-platform         | Research;Industry                  | Frameworks/Middle<br>ware             | Software                          | 8     | 8           | researchers, software<br>developers                               |
| Compilers                         | Databases in Ethereum Nodes                                                                               | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cross-platform         | Research;Industry                  | Libraries                             | Data                              | 7     | 6           | software developers,<br>researchers                               |
| Compilers                         | Bend, don't Break. Modern Techniques for Flexibility in Strongly Typed Functional Programming             | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 20  | Talk           | Cross-platform         | Community;Research                 | Libraries                             | Software                          | 7     | 5           | software developers,<br>researchers                               |
| Compilers                         | PeerDAS: Erasure Coding When You Can't Trust the Storage                                                  | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 20  | Talk           | Cross-platform         | Research;Industry                  | Libraries                             | Data                              | 8     | 8           | researchers, software<br>developers                               |
| Compilers                         | Benchmarking can be hard, here's why we're still creating a new one for OCaml                             | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cross-platform         | Research                           | Libraries                             | Software                          | 5     | 4           | researchers, software<br>developers                               |
| Compilers                         | What LLVM Does Differently When The Target Is CPU VS GPU                                                  | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cross-platform         | Industry;Research                  | Frameworks/Middle<br>ware             | Software                          | 8     | 6           | software developers,<br>researchers                               |
| Compilers                         | PythonBPF - now with loops, custom funcs, and more maps!                                                  | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 15  | Talk           | Cross-platform         | Community;Industry                 | Kernel;Libraries                      | Software                          | 7     | 7           | software developers,<br>hackers                                   |
| Compilers                         | LGTM Is Not A Theorem                                                                                     | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cross-platform         | Industry                           | Application                           | FOSS<br>Philosophy/Princ<br>iples | 4     | 3           | software developers,<br>professionals                             |
| Compilers                         | One Language, All GPU: How Julia Targets different GPU backends from a Single Kernel.                     | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 15  | Lightning Talk | Cross-platform         | Research;Community                 | Frameworks/Middle<br>ware             | Software                          | 7     | 7           | researchers, software<br>developers                               |
| Compilers                         | Rust Is Not an HDL, So I Built a Tiny One.                                                                | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 20  | Talk           | Desktop                | Community                          | Frameworks/Middle<br>ware             | Software                          | 6     | 6           | software developers,<br>hackers                                   |
| Compilers                         | Formal Verification: An Introduction through Elementary Arithmetic                                        | Day 1 (26 Sep)<br>Hall 3 · 14:00 | 20  | Talk           | Cross-platform         | Education/Academia;Re<br>search    | Application                           | Software                          | 5     | 5           | graduate/master students,<br>researchers,<br>educators/professors |
| Design                            | Forkable design                                                                                           | Day 2 (27 Sep)<br>Room 1 · 10:00 | 20  | Talk           | Web;Cross-<br>platform | Industry                           | Application                           | Software                          | 5     | 5           | designers, software<br>developers, business<br>people             |
| Design                            | Shaili Sanchaya: Reviving Indic Typography from Archival Metadata                                         | Day 2 (27 Sep)<br>Room 1 · 10:00 | 20  | Talk           | Web                    | Community;Not-for-Profit<br>Sector | Application                           | FOSS<br>Philosophy/Princ<br>iples | 7     | 8           | designers, researchers,<br>educators/professors                   |
| Design                            | Designing for Reality: Where UI Meets the Physical World                                                  | Day 2 (27 Sep)<br>Room 1 · 10:00 | 20  | Talk           | Cross-platform         | Industry                           | Application                           | Software                          | 4     | 4           | designers                                                         |
| Design                            | Creating Design Systems in p5JS to accelerate Content Creation                                            | Day 2 (27 Sep)<br>Room 1 · 10:00 | 20  | Talk           | Web                    | Industry;Community                 | Application                           | Software                          | 5     | 6           | designers, software<br>developers, business<br>people             |
| Design                            | KaiOS Design Notes: UX observations from a feature phone                                                  | Day 2 (27 Sep)<br>Room 1 · 10:00 | 20  | Talk           | Mobile                 | Community                          | Application                           | Software                          | 5     | 6           | designers, hackers                                                |
| Design                            | Making Desktops Expressive: Introducing Bézier Curves to GTK                                              | Day 2 (27 Sep)<br>Room 1 · 10:00 | 20  | Talk           | Desktop                | Community                          | Frameworks/Middle<br>ware;Application | Software                          | 6     | 6           | designers, software<br>developers                                 |
| Design                            | Sound Is All You Need: From Good UX to Great UX                                                           | Day 2 (27 Sep)<br>Room 1 · 10:00 | 20  | Talk           | Cross-platform         | Industry                           | Application                           | Software                          | 4     | 5           | designers                                                         |
| Documentation & Technical Writing | Your first open source contribution does not have to be code                                              | Day 1 (26 Sep)<br>Room 1 · 10:00 | 10  | Lightning Talk | Web                    | Community;Not-for-Profit<br>Sector | Application                           | FOSS<br>Philosophy/Princ<br>iples | 3     | 4           | hackers, social workers,<br>graduate/master students              |
| Documentation & Technical Writing | Why Documentation Shouldn't Feel Like Plain Text                                                          | Day 1 (26 Sep)<br>Room 1 · 10:00 | 10  | Lightning Talk | Cross-platform         | Community;Research                 | Frameworks/Middle<br>ware             | Software                          | 6     | 7           | software developers,<br>researchers                               |
| Documentation & Technical Writing | Documentation That Drives OSS Adoption: Lessons From Bruno                                                | Day 1 (26 Sep)<br>Room 1 · 10:00 | 20  | Talk           | Web                    | Industry;Community                 | Application                           | FOSS<br>Philosophy/Princ<br>iples | 5     | 4           | software developers,<br>professionals, business<br>people         |
| Documentation & Technical Writing | Ctrl+F for 34 Years of India's GovTech (Building a Public Commons for India's Digital Governance History) | Day 1 (26 Sep)<br>Room 1 · 10:00 | 10  | Lightning Talk | Web                    | Government;Legal/Polic<br>y        | Application                           | Data                              | 5     | 7           | researchers, government<br>employees, social<br>workers           |

| Track                             | Session                                                                                                       | When                             | Min | Format           | Platform       | Domain                                   | Stack Level          | Category                   | Depth | Novelt<br>y | Audience                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------|-----|------------------|----------------|------------------------------------------|----------------------|----------------------------|-------|-------------|--------------------------------------------------------------------------|
| Documentation & Technical Writing | Developer Experience is the New Open Source Contribution. Here's Why                                          | Day 1 (26 Sep)<br>Room 1 · 10:00 | 20  | Talk             | Cross-platform | Industry;Community                       | Application          | FOSS Philosophy/Principles | 5     | 5           | software developers, professionals                                       |
| Documentation & Technical Writing | Refactoring documentation without breaking it                                                                 | Day 1 (26 Sep)<br>Room 1 · 10:00 | 10  | Lightning Talk   | Web            | Community                                | Application          | Software                   | 4     | 4           | software developers, hackers                                             |
| Documentation & Technical Writing | Changelogs Are Not Enough: Writing Community Memory for Kubernetes                                            | Day 1 (26 Sep)<br>Room 1 · 10:00 | 10  | Lightning Talk   | Web            | Community;Industry                       | Application          | FOSS Philosophy/Principles | 4     | 5           | software developers, professionals                                       |
| Documentation & Technical Writing | Writing Docs for Two Readers: Humans and AI Agents                                                            | Day 1 (26 Sep)<br>Room 1 · 10:00 | 20  | Talk             | Web            | Industry;Community                       | Application          | FOSS Philosophy/Principles | 6     | 8           | software developers, professionals                                       |
| Documentation & Technical Writing | Python Now Speaks Tamil: Getting Official Docs Translated for a Mother Tongue                                 | Day 1 (26 Sep)<br>Room 1 · 10:00 | 10  | Lightning Talk   | Cross-platform | Education/Academia;Community             | Application          | FOSS Philosophy/Principles | 4     | 7           | educators/professors, graduate/master students, social workers           |
| Documentation & Technical Writing | 404: Empathy Not Found                                                                                        | Day 1 (26 Sep)<br>Room 1 · 10:00 | 20  | Talk             | Cross-platform | Community;Industry                       | Application          | FOSS Philosophy/Principles | 4     | 5           | software developers, professionals                                       |
| Documentation & Technical Writing | Printable, Downloadable, Forever: An Ode to Offline Documentation                                             | Day 1 (26 Sep)<br>Room 1 · 10:00 | 10  | Lightning Talk   | Cross-platform | Community                                | Application          | FOSS Philosophy/Principles | 3     | 4           | software developers, hackers                                             |
| General Track                     | From Pull Requests to Patients: Building Open Healthcare Infrastructure in the Open                           | Day 1 (26 Sep)<br>Hall 1 · 9:30  | 30  | Talk             | Web            | Not-for-Profit Sector;Government         | Application          | Software                   | 7     | 6           | software developers, professionals, social workers, government employees |
| General Track                     | From Institution to Community: Building COSH and a Culture of Open Source                                     | Day 1 (26 Sep)<br>Hall 1 · 9:30  | 30  | Talk             | Web            | Education/Academia                       | Application          | FOSS Philosophy/Principles | 4     | 5           | educators/professors, graduate/master students                           |
| General Track                     | Textbooks that run: interactive, zero-install CS education in the browser                                     | Day 1 (26 Sep)<br>Hall 1 · 9:30  | 30  | Talk             | Web            | Education/Academia                       | Application          | Software                   | 7     | 8           | educators/professors, graduate/master students, software developers      |
| General Track                     | Tracking-free and hackable network cameras - powered by libre software                                        | Day 1 (26 Sep)<br>Hall 1 · 9:30  | 30  | Talk             | Embedded       | Community                                | Firmware;Application | Hardware                   | 6     | 7           | hackers, software developers                                             |
| General Track                     | Fork Around and Find Out: How I Built a GNOME Extension Used by 200K+ People                                  | Day 1 (26 Sep)<br>Hall 1 · 9:30  | 15  | Lightning Talk   | Desktop        | Community                                | Application          | FOSS Philosophy/Principles | 4     | 4           | hackers, software developers                                             |
| General Track                     | It takes a village to raise a repo                                                                            | Day 1 (26 Sep)<br>Hall 1 · 9:30  | 15  | Lightning Talk   | Cross-platform | Community;Not-for-Profit Sector          | Application          | FOSS Philosophy/Principles | 3     | 3           | social workers, software developers                                      |
| General Track                     | Telling India's story through data                                                                            | Day 1 (26 Sep)<br>Hall 1 · 9:30  | 45  | Invited Talk     | Web            | Research;Community                       | Application          | Data                       | 6     | 6           | researchers, government employees, social workers                        |
| General Track                     | From DPI to People's Digital Infrastructure: Reclaiming Digital Infrastructures for Democratic Accountability | Day 1 (26 Sep)<br>Hall 1 · 14:30 | 60  | Panel Discussion | Web            | Government;Community ;Legal/Policy       | Application          | FOSS Philosophy/Principles | 5     | 6           | government employees, social workers, researchers, professionals         |
| General Track                     | Building OpenSpeaks tools for language documentation and archiving                                            | Day 1 (26 Sep)<br>Hall 1 · 14:30 | 30  | Talk             | Desktop;Web    | Research;Community;Not-for-Profit Sector | Application          | Software                   | 6     | 7           | researchers, social workers, software developers                         |
| General Track                     | Open Code, Curated Data: Building Open Water Intelligence for Indian Cities                                   | Day 1 (26 Sep)<br>Hall 1 · 14:30 | 30  | Talk             | Web            | Government;Community                     | Application          | Data                       | 7     | 7           | software developers, government employees, researchers                   |
| General Track                     | Building Zasper: Inception to Growth                                                                          | Day 1 (26 Sep)<br>Hall 2 · 9:45  | 30  | Talk             | Desktop        | Community;Research                       | Application          | Software                   | 6     | 5           | software developers, researchers                                         |
| General Track                     | Repub: Launching the Open Post-Processing Workflow for Digital Archiving in India                             | Day 1 (26 Sep)<br>Hall 2 · 9:45  | 30  | Talk             | Web;Desktop    | Not-for-Profit Sector;Community          | Application          | Data                       | 6     | 7           | social workers, researchers, software developers                         |

| Track         | Session                                                                                            | When                             | Min | Format                | Platform         | Domain                           | Stack Level                       | Category                   | Depth | Novelty | Audience                                                            |
|---------------|----------------------------------------------------------------------------------------------------|----------------------------------|-----|-----------------------|------------------|----------------------------------|-----------------------------------|----------------------------|-------|---------|---------------------------------------------------------------------|
| General Track | How to Build Long-Lasting Tech Communities That Thrive — A Playbook                                | Day 1 (26 Sep)<br>Hall 2 · 9:45  | 30  | Talk                  | Cross-platform   | Community                        | Application                       | FOSS Philosophy/Principles | 4     | 4       | community organizers, software developers, business people          |
| General Track | Optimizing Performance of GUIs                                                                     | Day 1 (26 Sep)<br>Hall 2 · 9:45  | 30  | Talk                  | Embedded;Desktop | Industry                         | Frameworks/Middleware             | Software                   | 7     | 6       | software developers, professionals                                  |
| General Track | FQ-PIE: Contributing to Open Standards by Contributing to FOSS                                     | Day 1 (26 Sep)<br>Hall 2 · 9:45  | 15  | Lightning Talk        | Web              | Research;Community               | Frameworks/Middleware             | Software                   | 6     | 6       | researchers, software developers                                    |
| General Track | Mapping India in seconds, for one and all                                                          | Day 1 (26 Sep)<br>Hall 2 · 9:45  | 15  | Lightning Talk        | Web              | Government;Research              | Application                       | Data                       | 6     | 7       | government employees, researchers, software developers              |
| General Track | FOSS for the Forgotten Middle: Open-Sourcing Finance Software for India's 50,000 Microgyms         | Day 1 (26 Sep)<br>Hall 2 · 14:00 | 30  | Talk                  | Web              | Industry                         | Application                       | Software                   | 5     | 4       | business people, software developers                                |
| General Track | Navigating the FOSS Contribution Journey                                                           | Day 1 (26 Sep)<br>Hall 2 · 14:00 | 30  | Talk                  | Cross-platform   | Community                        | Application                       | FOSS Philosophy/Principles | 3     | 3       | graduate/master students, software developers                       |
| General Track | How FOSS and AI are Changing Government Workflows at NHAI                                          | Day 1 (26 Sep)<br>Hall 2 · 14:00 | 30  | Talk                  | Web              | Government                       | Application                       | Software                   | 6     | 7       | government employees, software developers, professionals            |
| General Track | Offline-first mobile apps on Frappe: a metadata-driven Flutter SDK with server-side control        | Day 1 (26 Sep)<br>Hall 2 · 14:00 | 30  | Talk                  | Mobile           | Industry                         | Frameworks/Middleware;Application | Software                   | 8     | 7       | software developers, professionals                                  |
| General Track | Physics at 60FPS                                                                                   | Day 1 (26 Sep)<br>Hall 2 · 14:00 | 30  | Talk                  | Cross-platform   | Community;Industry               | Libraries                         | Software                   | 7     | 6       | software developers, professionals                                  |
| General Track | Do Open Source Companies Pay Competitively?                                                        | Day 1 (26 Sep)<br>Room 2 · 11:00 | 60  | Birds of Feather(BoF) | Web              | Industry                         | Application                       | FOSS Philosophy/Principles | 3     | 4       | software developers, business people, professionals                 |
| General Track | FOSS and Co-operatives                                                                             | Day 1 (26 Sep)<br>Room 2 · 14:30 | 60  | Birds of Feather(BoF) | Cross-platform   | Legal/Policy;Community           | Application                       | FOSS Philosophy/Principles | 4     | 5       | business people, software developers, social workers                |
| General Track | FOSS4Good - Techies in the social sector                                                           | Day 1 (26 Sep)<br>Room 3 · 11:00 | 60  | Birds of Feather(BoF) | Cross-platform   | Not-for-Profit Sector;Community  | Application                       | FOSS Philosophy/Principles | 3     | 4       | social workers, software developers                                 |
| General Track | FOSS in the Public Consciousness                                                                   | Day 1 (26 Sep)<br>Room 3 · 14:30 | 60  | Birds of Feather(BoF) | Cross-platform   | Community;Legal/Policy           | Application                       | FOSS Philosophy/Principles | 3     | 4       | software developers, social workers, business people                |
| General Track | How India's Largest G2C Platform Migrated to FOSS and Broke Free from Oracle                       | Day 2 (27 Sep)<br>Hall 1 · 9:45  | 30  | Talk                  | Web              | Government                       | Application                       | Software                   | 7     | 6       | government employees, software developers, professionals            |
| General Track | How exactly can we teach students the right FOSS etiquette?                                        | Day 2 (27 Sep)<br>Hall 1 · 9:45  | 30  | Talk                  | Cross-platform   | Education/Academia;Community     | Application                       | FOSS Philosophy/Principles | 4     | 5       | educators/professors, graduate/master students, software developers |
| General Track | Seqout: search engine for the world's genomic data                                                 | Day 2 (27 Sep)<br>Hall 1 · 9:45  | 30  | Talk                  | Web              | Research                         | Application                       | Data                       | 8     | 7       | researchers, software developers                                    |
| General Track | How we build Community Projects from the Community for the Community                               | Day 2 (27 Sep)<br>Hall 1 · 9:45  | 30  | Talk                  | Web              | Community                        | Application                       | FOSS Philosophy/Principles | 4     | 4       | software developers, hackers                                        |
| General Track | What can we do about open weight models enabling non consensual intimate imagery?                  | Day 2 (27 Sep)<br>Hall 1 · 9:45  | 15  | Lightning Talk        | Web              | Legal/Policy;Research            | Application                       | FOSS Philosophy/Principles | 5     | 6       | researchers, government employees, social workers                   |
| General Track | Opening the Archives: Building Digital Infrastructure for Public Participation in Archival Records | Day 2 (27 Sep)<br>Hall 1 · 9:45  | 15  | Lightning Talk        | Web              | Government;Not-for-Profit Sector | Application                       | Data                       | 6     | 6       | researchers, government employees, social workers                   |
| General Track | Flexibility is complexity in disguise: Building frappe-ui for 10+ Products                         | Day 2 (27 Sep)<br>Hall 1 · 9:45  | 30  | Talk                  | Web              | Industry                         | Frameworks/Middleware;Libraries   | Software                   | 7     | 6       | software developers, professionals                                  |

| Track         | Session                                                                                                           | When                             | Min | Format                   | Platform       | Domain                                        | Stack Level                | Category                          | Depth | Novelt<br>y | Audience                                                                                    |
|---------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------|-----|--------------------------|----------------|-----------------------------------------------|----------------------------|-----------------------------------|-------|-------------|---------------------------------------------------------------------------------------------|
| General Track | The future of FOSS, SWE and Technical Education                                                                   | Day 2 (27 Sep)<br>Hall 1 · 14:30 | 60  | Panel Discussion         | Cross-platform | Education/Academia;Ind<br>ustry               | Application                | FOSS<br>Philosophy/Princ<br>iples | 4     | 6           | educators/professors,<br>software developers,<br>professionals,<br>graduate/master students |
| General Track | Don't Model Every Leaf: The Magic of Blender<br>Geometry Nodes                                                    | Day 2 (27 Sep)<br>Hall 2 · 9:45  | 15  | Lightning Talk           | Desktop        | Community                                     | Application                | Software                          | 4     | 4           | designers, hackers                                                                          |
| General Track | Property-based testing for UIs                                                                                    | Day 2 (27 Sep)<br>Hall 2 · 9:45  | 15  | Lightning Talk           | Mobile;Web     | Industry;Community                            | Application                | Software                          | 6     | 6           | software developers,<br>professionals                                                       |
| General Track | Surviving as an Open Source Startup : Xeneva                                                                      | Day 2 (27 Sep)<br>Hall 2 · 9:45  | 30  | Talk                     | Embedded       | Industry                                      | Operating<br>System;Kernel | Hardware                          | 7     | 8           | software developers,<br>hackers, professionals                                              |
| General Track | OpenTCAD: A FOSS Python Pipeline for<br>Semiconductor Process and Device Simulation                               | Day 2 (27 Sep)<br>Hall 2 · 9:45  | 15  | Lightning Talk           | Desktop        | Research;Industry                             | Application                | Hardware                          | 8     | 8           | researchers, software<br>developers                                                         |
| General Track | Open, decentralized language assessment software<br>for the public school system                                  | Day 2 (27 Sep)<br>Hall 2 · 9:45  | 15  | Lightning Talk           | Mobile;Web     | Education/Academia;Not<br>-for-Profit Sector  | Application                | Software                          | 6     | 7           | educators/professors,<br>social workers, software<br>developers                             |
| General Track | Computationalizing landscapes: The CoRE stack data<br>structure for social-ecological diagnostics and<br>planning | Day 2 (27 Sep)<br>Hall 2 · 9:45  | 30  | Talk                     | Web            | Research;Government;N<br>ot-for-Profit Sector | Application                | Data                              | 7     | 7           | researchers, government<br>employees, social<br>workers                                     |
| General Track | Bad Internet, Good Software                                                                                       | Day 2 (27 Sep)<br>Hall 2 · 9:45  | 15  | Lightning Talk           | Desktop        | Community                                     | Application                | Software                          | 4     | 4           | hackers, software<br>developers                                                             |
| General Track | Open Standards in XR - A current overview of<br>OpenXR and WebXR                                                  | Day 2 (27 Sep)<br>Hall 2 · 9:45  | 15  | Lightning Talk           | Cross-platform | Industry;Community                            | Frameworks/Middle<br>ware  | Software                          | 6     | 5           | software developers,<br>professionals                                                       |
| General Track | Freedom to code - Community's role in tech Policy                                                                 | Day 2 (27 Sep)<br>Hall 2 · 9:45  | 30  | Talk                     | Cross-platform | Legal/Policy                                  | Application                | FOSS<br>Philosophy/Princ<br>iples | 5     | 4           | business people,<br>government employees,<br>social workers                                 |
| General Track | Software Patents - AI and the Future of Open<br>Innovation                                                        | Day 2 (27 Sep)<br>Hall 2 · 14:00 | 15  | Lightning Talk           | Web            | Legal/Policy                                  | Application                | FOSS<br>Philosophy/Princ<br>iples | 5     | 5           | business people, software<br>developers, government<br>employees                            |
| General Track | Calibrate: A framework for continuously evaluating and<br>improving AI agents within non-profits                  | Day 2 (27 Sep)<br>Hall 2 · 14:00 | 30  | Talk                     | Cross-platform | Not-for-Profit<br>Sector;Industry             | Frameworks/Middle<br>ware  | Software                          | 7     | 7           | software developers,<br>professionals, social<br>workers                                    |
| General Track | Being Bayesian: Stress-testing India's Loudest Stories<br>with FOSS & Open Data                                   | Day 2 (27 Sep)<br>Hall 2 · 14:00 | 30  | Talk                     | Web            | Research;Community                            | Application                | Data                              | 6     | 6           | researchers, software<br>developers, social<br>workers                                      |
| General Track | Building Open Source Digital Infrastructure for a<br>University                                                   | Day 2 (27 Sep)<br>Hall 2 · 14:00 | 15  | Lightning Talk           | Web            | Education/Academia                            | Application                | Software                          | 5     | 4           | graduate/master students,<br>software developers,<br>educators/professors                   |
| General Track | Judiciary Entity Map: How open source can make<br>India's courts legible                                          | Day 2 (27 Sep)<br>Hall 2 · 14:00 | 30  | Talk                     | Web            | Legal/Policy;Governmen<br>t                   | Application                | Data                              | 7     | 7           | researchers, government<br>employees, social<br>workers                                     |
| General Track | Can We Build a Copyleft License for Community Data<br>in the Age of AI                                            | Day 2 (27 Sep)<br>Room 2 · 10:30 | 60  | Birds of<br>Feather(BoF) | Web;Data       | Legal/Policy;Community                        | Application                | FOSS<br>Philosophy/Princ<br>iples | 5     | 7           | researchers, business<br>people, social workers                                             |
| General Track | Micro-granting in India                                                                                           | Day 2 (27 Sep)<br>Room 2 · 11:30 | 60  | Birds of<br>Feather(BoF) | Cross-platform | Not-for-Profit<br>Sector;Community            | Application                | FOSS<br>Philosophy/Princ<br>iples | 2     | 3           | social workers, business<br>people                                                          |
| General Track | Community-Driven or Community-Branded? Who Gets<br>to Shape Community Events?                                     | Day 2 (27 Sep)<br>Room 2 · 14:30 | 60  | Birds of<br>Feather(BoF) | Cross-platform | Community                                     | Application                | FOSS<br>Philosophy/Princ<br>iples | 2     | 3           | software developers,<br>social workers                                                      |
| General Track | Community-Centred Technology                                                                                      | Day 2 (27 Sep)<br>Room 3 · 10:30 | 60  | Birds of<br>Feather(BoF) | Web;Embedded   | Not-for-Profit<br>Sector;Community            | Application                | FOSS<br>Philosophy/Princ<br>iples | 4     | 5           | social workers,<br>researchers                                                              |
| General Track | How do we scale and sustain AI adoption without<br>burning the planet?                                            | Day 2 (27 Sep)<br>Room 3 · 11:30 | 60  | Birds of<br>Feather(BoF) | Cross-platform | Legal/Policy;Not-for-<br>Profit Sector        | Application                | FOSS<br>Philosophy/Princ<br>iples | 4     | 5           | business people,<br>government employees,<br>social workers                                 |

| Track         | Session                                                                                       | When                             | Min | Format                | Platform          | Domain                      | Stack Level               | Category                   | Depth | Novelty | Audience                                                             |
|---------------|-----------------------------------------------------------------------------------------------|----------------------------------|-----|-----------------------|-------------------|-----------------------------|---------------------------|----------------------------|-------|---------|----------------------------------------------------------------------|
| General Track | United by (Free) F&B – Food & Books                                                           | Day 2 (27 Sep)<br>Room 3 · 12:30 | 45  | Birds of Feather(BoF) | Cross-platform    | Community                   | Application               | FOSS Philosophy/Principles | 1     | 1       | community                                                            |
| General Track | FOSS for Justice — What can Open Source do for India's Courts?                                | Day 2 (27 Sep)<br>Room 3 · 14:30 | 60  | Birds of Feather(BoF) | Web               | Legal/Policy;Government     | Application               | Data                       | 4     | 5       | government employees, social workers, software developers            |
| Hardware      | Minnow: An Open Platform for Ground Swarm Robotics                                            | Day 1 (26 Sep)<br>Hall 3 · 10:00 | 25  | Talk                  | Embedded/Hardware | Research;Education/Academia | Hardware;Firmware         | Hardware                   | 6     | 7       | researchers, hackers, graduate/master students, educators/professors |
| Hardware      | Build your own open source keyboard with ZMK/QMK                                              | Day 1 (26 Sep)<br>Hall 3 · 10:00 | 30  | Talk                  | Embedded/Hardware | Community                   | Firmware                  | Hardware                   | 5     | 4       | hackers, software developers                                         |
| Hardware      | Building Open Source Music Hardware in India - Tarab Instruments                              | Day 1 (26 Sep)<br>Hall 3 · 10:00 | 30  | Talk                  | Embedded/Hardware | Community;Industry          | Hardware                  | Hardware                   | 5     | 6       | hackers, professionals                                               |
| Hardware      | CNC4Everyone - How we built a CNC machine for every classroom, college and makerspace         | Day 1 (26 Sep)<br>Hall 3 · 10:00 | 30  | Talk                  | Embedded/Hardware | Education/Academia          | Hardware;Firmware         | Hardware                   | 6     | 6       | educators/professors, hackers, graduate/master students              |
| Hardware      | My zero-to-hero journey deploying a fiber-optic and wireless community mesh                   | Day 1 (26 Sep)<br>Hall 3 · 10:00 | 30  | Talk                  | Embedded/Hardware | Community                   | Hardware;Application      | Hardware                   | 5     | 7       | hackers, social workers, community                                   |
| Hardware      | HackerFab IITB: democratizing semiconductor fabrication                                       | Day 1 (26 Sep)<br>Hall 3 · 10:00 | 30  | Talk                  | Hardware          | Research;Education/Academia | Hardware                  | Hardware                   | 8     | 9       | researchers, graduate/master students, hackers                       |
| RTOS          | Beyond Hardware Description: Using Device Tree for System-Level Configuration in Zephyr RTOS  | Day 2 (27 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk        | Embedded          | Industry                    | Firmware;Operating System | Software                   | 7     | 5       | software developers, professionals                                   |
| RTOS          | From Display Controller to Panel: A Modular Display Pipeline Architecture for Zephyr          | Day 2 (27 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk        | Embedded          | Industry                    | Firmware;Operating System | Software                   | 8     | 7       | software developers, professionals                                   |
| RTOS          | From Reset to main(): Porting Infineon TriCore AURIX to Zephyr                                | Day 2 (27 Sep)<br>Room 1 · 14:00 | 25  | Talk                  | Embedded          | Industry                    | Firmware;Operating System | Software                   | 9     | 8       | software developers, professionals, researchers                      |
| RTOS          | Real-Time Video Pipelines on Embedded SoCs: A FreeRTOS Case Study                             | Day 2 (27 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk        | Embedded          | Industry                    | Firmware                  | Software                   | 8     | 6       | software developers, professionals                                   |
| RTOS          | Tackling core lifecycle management in heterogeneous SoCs                                      | Day 2 (27 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk        | Embedded          | Industry                    | Firmware;Operating System | Software                   | 8     | 7       | software developers, professionals                                   |
| RTOS          | Loops to real time                                                                            | Day 2 (27 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk        | Embedded          | Education/Academia          | Firmware;Operating System | Software                   | 3     | 2       | graduate/master students, hackers, software developers               |
| RTOS          | Indigenous, High-Performance, Safety-Critical RTOS                                            | Day 2 (27 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk        | Embedded          | Government;Industry         | Operating System;Kernel   | Software                   | 8     | 8       | professionals, researchers, government employees                     |
| RTOS          | From Architecture to Tasks: Designing Real-Time MCU Firmware in Heterogeneous APU-MCU Systems | Day 2 (27 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk        | Embedded          | Industry                    | Firmware                  | Software                   | 8     | 6       | software developers, professionals                                   |
| RTOS          | Deterministic Vision: Offloading Real-Time Camera Pipelines to FreeRTOS on Heterogeneous SoCs | Day 2 (27 Sep)<br>Room 1 · 14:00 | 15  | Lightning Talk        | Embedded          | Industry                    | Firmware                  | Software                   | 8     | 7       | software developers, professionals                                   |
| RTOS          | RTL to Zephyr RTOS: Lightweight-Crypto Acceleration on Open RISC-V                            | Day 2 (27 Sep)<br>Room 1 · 14:00 | 15  | Talk                  | Embedded;Hardware | Research;Industry           | Hardware;Firmware         | Hardware                   | 9     | 9       | researchers, software developers, hackers                            |
| Security      | Reverse Engineering the PAN QR Code                                                           | Day 2 (27 Sep)<br>Hall 3 · 10:00 | 15  | Lightning Talk        | Mobile;Web        | Government;Community        | Application               | Software                   | 6     | 7       | hackers, software developers, government employees                   |
| Security      | When NO_PROXY Lies: Finding an IPv4-Mapped IPv6 Patch Bypass in Axios                         | Day 2 (27 Sep)<br>Hall 3 · 10:00 | 15  | Lightning Talk        | Web               | Industry;Community          | Libraries                 | Software                   | 7     | 6       | software developers, professionals                                   |
| Security      | Extracting AES Keys from Embedded Devices Using Open Source Hardware                          | Day 2 (27 Sep)<br>Hall 3 · 10:00 | 20  | Lightning Talk        | Embedded          | Research;Industry           | Hardware;Firmware         | Hardware                   | 7     | 6       | hackers, researchers, professionals                                  |
| Security      | The Art of Userspace Sandboxing on Linux                                                      | Day 2 (27 Sep)<br>Hall 3 · 10:00 | 25  | Talk                  | Desktop           | Industry;Community          | Kernel;Operating System   | Software                   | 8     | 6       | software developers, professionals, hackers                          |

| Track                          | Session                                                                                                  | When                             | Min | Format         | Platform               | Domain                                       | Stack Level                        | Category                   | Depth | Novelty | Audience                                                   |
|--------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------|-----|----------------|------------------------|----------------------------------------------|------------------------------------|----------------------------|-------|---------|------------------------------------------------------------|
| Security                       | Extending Capa for Malicious Script Analysis                                                             | Day 2 (27 Sep)<br>Hall 3 · 10:00 | 15  | Lightning Talk | Desktop                | Community;Research                           | Libraries;Application              | Software                   | 7     | 6       | software developers, researchers                           |
| Security                       | Keeping an Eye on your Eye: Disassembling and Reassembling Smart Camera Hardware to breach YOUR Security | Day 2 (27 Sep)<br>Hall 3 · 10:00 | 30  | Talk           | Embedded               | Community                                    | Hardware;Firmware                  | Hardware                   | 8     | 7       | hackers, professionals                                     |
| Security                       | Hacking India's Largest Exam System                                                                      | Day 2 (27 Sep)<br>Hall 3 · 10:00 | 15  | Lightning Talk | Web                    | Government;Community                         | Application                        | Software                   | 5     | 6       | hackers, government employees, social workers              |
| Security                       | Papers, Please: A First Look at Age Verification on the Web                                              | Day 2 (27 Sep)<br>Hall 3 · 10:00 | 25  | Talk           | Web                    | Research;Legal/Policy                        | Application                        | Data                       | 7     | 7       | researchers, government employees, professionals           |
| Security                       | A plan to sort out this CVE mess                                                                         | Day 2 (27 Sep)<br>Hall 3 · 10:00 | 15  | Lightning Talk | Web                    | Community;Industry                           | Application                        | Data                       | 6     | 5       | software developers, professionals, government employees   |
| Workshops (Pre-Conference Day) | Getting Started with Qt                                                                                  | Pre-Conf. Day<br>Workshop Venue  | 120 | Workshop       | Desktop;Cross-platform | Community;Education/Academia                 | Frameworks/Middle ware;Application | Software                   | 5     | 3       | software developers, hackers, graduate/master students     |
| Workshops (Pre-Conference Day) | Making music with strudel.cc                                                                             | Pre-Conf. Day<br>Workshop Venue  | 120 | Workshop       | Web                    | Education/Academia;Community                 | Application                        | Software                   | 3     | 6       | hackers, designers, graduate/master students               |
| Workshops (Pre-Conference Day) | Digital Security for Journalists, Activists, and Everyday Internet Users: A Hands-On FOSS Workshop       | Pre-Conf. Day<br>Workshop Venue  | 120 | Workshop       | Cross-platform         | Not-for-Profit Sector;Community;Legal/Policy | Application                        | FOSS Philosophy/Principles | 4     | 4       | social workers, hackers, professionals                     |
| Workshops (Pre-Conference Day) | Fun and Profit with OCaml                                                                                | Pre-Conf. Day<br>Workshop Venue  | 120 | Workshop       | Cross-platform         | Education/Academia;Research                  | Libraries;Application              | Software                   | 6     | 5       | graduate/master students, software developers, researchers |
| Workshops (Pre-Conference Day) | Art - meet Math - meet Code                                                                              | Pre-Conf. Day<br>Workshop Venue  | 120 | Workshop       | Web                    | Education/Academia;Community                 | Application                        | Software                   | 4     | 6       | designers, educators/professors, hackers                   |
| Workshops (Pre-Conference Day) | Building AI agents end-to-end: A hands-on workshop                                                       | Pre-Conf. Day<br>Workshop Venue  | 120 | Workshop       | Cross-platform         | Industry;Community                           | Frameworks/Middle ware;Application | Software                   | 7     | 7       | software developers, professionals                         |
| Workshops (Pre-Conference Day) | Adding places to OpenStreetMap (outdoor workshop)                                                        | Pre-Conf. Day<br>Workshop Venue  | 120 | Workshop       | Mobile                 | Community;Not-for-Profit Sector              | Application                        | Data                       | 3     | 3       | hackers, social workers                                    |
| Workshops (Pre-Conference Day) | Self-Hosting 101: A real-time self-hosting workshop                                                      | Pre-Conf. Day<br>Workshop Venue  | 120 | Workshop       | Cloud/Server           | Community                                    | Application;Operating System       | Software                   | 5     | 4       | hackers, software developers                               |

## Note on Methodology & Categorization

This report was produced by an AI assistant (Claude, by Anthropic) working through the conference's own source files, not by the IndiaFOSS 2026 programme committee. It is intended as an analytical starting point for programme review, not as an authoritative or final judgement on any session, speaker, or track.

### How the session list was built

1. **Schedule extraction.** `if2026-schedule.ods` was parsed sheet-by-sheet (General Track plus the 8 devroom sheets, excluding the Food Breaks sheet). Day, hall and start-time values are only written on the first row of each schedule block in the source file, so they were forward-filled down subsequent rows within a block; the per-row `exclude` flag (General Track only) was applied per-row, not forward-filled, to avoid dropping unrelated sessions that merely followed an excluded one.
2. **Title matching.** Each session title was normalised (smart quotes, dashes and whitespace collapsed) and matched against `cfp-all.csv` to recover its CFP ID. All 129 main-programme sessions and all 8 pre-conference workshops matched an ID; nothing was matched by fuzzy/approximate title similarity.
3. **Proposal retrieval.** For every matched ID, the full public proposal page at `fossunited.org/c/indiafoss/2026/cfp/<ID>` was fetched and its "Session Description" and "Key Takeaways" fields were extracted as plain text. All scoring and tagging below was based on reading this original proposal text, not on the title alone.
4. **Workshops.** The 8 Workshop-type entries in `cfp-all.csv` do not appear in `if2026-schedule.ods` because they run on a separate pre-conference day; they were added to the dataset manually as a 10th track ("Workshops (Pre-Conference Day)") with an assumed uniform duration of 120 minutes each, per the stated ~2-hour format, since no individual start times were published for them.

### How each session was classified

Every one of the 137 sessions was individually read and tagged by the AI assistant against the seven dimensions defined in `categories.md`:

- **Platform, Domain, Level in Technology Stack** — assigned as one or more free-text tags per session (separated by " ; " in the data table), based on the technology and audience the proposal itself describes. A session can and often does carry more than one tag in each field.
- **Overall category** — a single primary label (Hardware / Software / Data / FOSS Philosophy & Principles) chosen to reflect the proposal's centre of gravity, not every topic it touches.
- **Depth of technical coverage (1–10)** and **Novelty vs. existing work (1–10)** — judgement calls made by the AI assistant from the proposal text alone (scope, prerequisites assumed, specificity of implementation detail, and how the author frames the work's relationship to prior art). These were *not* derived from any objective external signal (e.g. citation counts, GitHub stars, speaker seniority) and should be read as one analyst's relative ranking across the programme, not an absolute or verified score.
- **Suitability for audience** — one or more audience tags from the `categories.md` list, inferred from who the proposal explicitly addresses or clearly assumes as background.

#### Where this analysis is most likely to be wrong

- Depth and novelty scores are subjective single-pass estimates from proposal text, not from watching the actual delivered talk — a strong speaker can make a "5/10 novelty" topic land as fresh, and a weak delivery can flatten a "9/10" one.
- Classification was done from the CFP abstract only; last-minute scope changes, co-speakers, or slide content that diverges from the original proposal are not reflected.
- Multi-tag fields (platform, domain, stack level, audience) reflect the assistant's reading of emphasis, which is inherently approximate for sessions that genuinely span several areas.
- The dataset reflects a single snapshot of `if2026-schedule.ods` and `cfp-all.csv`; if the schedule changes after this report was generated, the numbers here will no longer match the live programme.

The full per-session table earlier in this report is provided precisely so that any individual classification can be checked against the linked original proposal and corrected by someone with direct knowledge of the talk.